When Does Segmentation Pay?

Segment Targeting versus Product Line Design in Conjoint Analysis

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Abstract

A common way to use conjoint data is to split consumers into segments and design one product per segment. I compare this strategy with a single product for everyone and with a product line chosen directly from individual-level preferences. The data are ratings of 20 messenger bags by 122 respondents. Latent class models keep improving fit up to eight segments, but holdout prediction stops improving after three, and only two segments are stable. When products are chosen on one half of the respondents and evaluated on the other, targeting latent class segments lowers revenue by 1.5-3.9% relative to one product for everyone, while an optimal product line raises it by 2.6-7.2%. In simulations calibrated to the data, targeting pays only when segments are well separated and the firm surveys several hundred people. Choosing to segment because BIC prefers more than one segment loses 4.5% of revenue on average; choosing by revenue on the firm's own sample loses less than 1%.

1 Introduction

Segmentation based on conjoint data usually has three steps: estimate preferences, group consumers with similar preferences, and design a product for each group (Wedel and Kamakura, 2000). The number of segments is usually chosen by a fit criterion such as BIC, and the revenue of the segment products is checked on the same respondents.

I ask when this approach increases revenue compared with designing a product line directly from individual preferences. The answer is not obvious for two reasons. A segmentation model can fit the data better without the segments being useful for pricing and design. And both the segments and the products are chosen from a finite sample, so a strategy that wins in sample can lose on new consumers.

I first use a ratings-based conjoint study of messenger bags and evaluate every strategy out of sample. I then simulate markets calibrated to the same data, vary the structure of heterogeneity and the sample size, and split the realized gain of each strategy into the gain with known preferences and the loss from estimation. Finally, I check whether the firm can tell from its own survey which strategy to use.

The main findings are as follows. In the data, fit, prediction and stability point to different numbers of segments, and hierarchical Bayes (HB) predicts better than any number of latent class (LC) segments. Out of sample, targeting segments loses money and a product line chosen on individual partworths gains. In the simulation, targeting pays only with well separated segments and large samples, and a decision to segment based on BIC costs 4.5% of revenue.

2 Related literature

Latent class models estimate segment membership and segment-level parameters jointly (Kamakura and Russell, 1989); Wedel and Kamakura (2000) review their use for segmentation and conjoint analysis. An alternative is to model heterogeneity with a continuous distribution and estimate it by hierarchical Bayes (Allenby and Rossi, 1998). Allenby et al. (1998) show that heterogeneity within segments stays large even when several mixture components are

allowed. Andrews et al. (2002) compare HB and finite mixture models in simulated conjoint data: the two predict holdout ratings about equally well, and individual-level OLS overfits.

These papers compare models by fit, prediction and parameter recovery. Product line design from conjoint estimates is studied separately as an optimization problem; Belloni et al. (2008) show that simple heuristics such as product swapping usually reach the optimal line. Elmachtoub and Grigas (2022) argue more generally that a model used for optimization should be judged by the decisions it produces.

This paper connects these two parts. It evaluates segmentation by the out-of-sample revenue of the products it leads to, not by fit, and it compares segment targeting with direct line design under the same demand model. It does not propose a new estimator.

3 Data

Each of 122 respondents rated the same 20 bags on a scale from 1 ("definitely would not buy") to 5 ("definitely would buy"). Bags differ in six attributes (Table 1), which gives 384 possible bags. The same respondents later rated the same bags offline; I use these ratings only in Section 8.

Table 1. Attributes and levels

Attribute	Levels
Exterior design	Black (base), reflective, colorful, blue
Size	Small (base), large
Price	\$120, \$140, \$160, \$180
Strap pad	No (base), yes
Water bottle pocket	No (base), yes
Interior	No dividers (base), file divider, padded laptop compartment

Notes: Price is coded from -0.5 (\$120) to 0.5 (\$180), so its coefficient is the change in rating from the lowest to the highest price. Other attributes are dummy coded.

The design is close to orthogonal: the largest correlation between columns of different attributes is 0.18. With 20 ratings and 10 parameters per person, individual estimates are noisy. About 14% of the rating variance comes from differences in respondents' mean ratings. One offline rating equals 2.5 and is kept as recorded.

4 Methods

Preferences. Ratings are linear in partworths, $r_{ij} = z_j'\beta_i + \varepsilon_{ij}$ with $\varepsilon_{ij} \sim N(0, \sigma^2)$, where z_j is the coded profile with an intercept. Pooled OLS sets $\beta_i = \beta$ and clusters standard errors by respondent. Individual OLS runs one regression per respondent. HB assumes $\beta_i \sim N(\mu, \Sigma)$ with a flat prior on μ, $\Sigma \sim IW(p+3, 0.1(p+3)I)$ and $\sigma^2 \sim IG(1,1)$, and is estimated by Gibbs sampling (6,000 draws, 2,000 burn-in).

Segments. In the LC model, respondent i belongs to segment k with probability π_k, and all of i's ratings come from that segment's partworths. To keep segments from simply separating generous and strict raters, I subtract each respondent's mean rating and the profile means, and maximize

$$L = \prod_{i=1}^{N} \sum_{k=1}^{K} \pi_k \prod_{j=1}^{J} \phi\left(r_{ij} - \bar{r}_i;\ (z_j - \bar{z})' b_k,\ \sigma_k^2\right) \tag{1}$$

by EM with 20 random starts. I report BIC with N equal to the number of respondents and ICL, which adds twice the entropy of the posterior memberships. As an alternative, I cluster standardized HB partworths with k-means. Stability

is the mean adjusted Rand index (ARI) between the full-sample partition and partitions from 50 bootstrap samples.

Demand and strategies. A consumer buys the offered bag with the highest predicted rating $u_{ij} = z_j'\beta_i$ if this rating is at least 3. For a set of bags $\mathscr{L}$,

$$R(\mathscr{L}) = \sum_i p_{j^*(i)} \mathbf{1}\{u_{ij^*(i)} \geq 3\}, \qquad j^*(i) = \arg\max_{j \in \mathscr{L}} u_{ij}. \tag{2}$$

I compare four strategies with K bags. *Mass market*: one bag that maximizes R. *Targeted segments*: each segment gets the bag that maximizes revenue among its members, and consumers can only buy their segment's bag. *Segment products, open*: the same bags, offered to everyone. *Optimal line*: the K bags that maximize R jointly, found by a greedy start and single-bag swaps; for $K = 2$ this matches full enumeration of all 73,536 pairs.

Evaluation. I use 10 repetitions of 2-fold cross-validation over respondents. On the training half the firm estimates HB and segments and chooses bags. Revenue is computed on the test half with those respondents' full-sample HB partworths, and test respondents are assigned to segments from their own ratings. The *realized gain* is revenue relative to a mass-market bag chosen the same way. The *potential gain* repeats the same choices with the test respondents' own partworths, as if the firm knew them. The folds overlap, so standard errors across folds are somewhat too small.

5 Results

5.1 Preferences

All attributes are significant (Table 2). Respondents dislike higher prices and the colorful exterior and like every functional feature. The spread across respondents is about as large as the mean: 93-96% of respondents value each functional feature positively, but by very different amounts. Respondents disagree most about the exterior.

Table 2. Partworths: pooled OLS and hierarchical Bayes

	Pooled OLS		Hierarchical Bayes		
	Coef.	(SE)	Mean	SD across respondents	Share positive
Intercept	2.05^{***}	(0.10)	2.05	0.90	1.00
Price ($120 → $180)	-0.67^{***}	(0.07)	−0.67	0.57	0.05
Exterior: reflective	-0.31^{***}	(0.07)	−0.31	0.57	0.22
Exterior: colorful	-1.06^{***}	(0.10)	−1.06	1.00	0.14
Exterior: blue	-0.22^{***}	(0.06)	−0.21	0.44	0.25
Size: large	0.27^{***}	(0.05)	0.27	0.46	0.78
Strap pad	0.51^{***}	(0.05)	0.51	0.47	0.93
Water bottle pocket	0.45^{***}	(0.04)	0.45	0.38	0.96
Interior: file divider	0.41^{***}	(0.05)	0.41	0.39	0.96
Interior: padded laptop	0.62^{***}	(0.06)	0.62	0.59	0.93

Notes: 122 respondents, 2,440 ratings. Standard errors clustered by respondent; $^{***}p <$ 0.01. HB: posterior mean of μ, square root of the posterior mean of the diagonal of Σ, and share of respondents with a positive posterior mean partworth.

To compare models, I hide 4 of the 20 bags for all respondents, estimate on the other 16 and predict the hidden ratings (10 splits). HB is best on both measures (Table 3). Individual OLS is barely better than pooled OLS because its estimates are noisy, as in Andrews et al. (2002).

Table 3. Prediction of hidden bags

Model	RMSE	Hit rate
Pooled OLS	1.112 (0.029)	0.731 (0.040)
Pooled OLS + respondent intercept	1.059 (0.032)	0.731 (0.040)
Individual OLS	1.091 (0.044)	0.762 (0.018)
Latent class, $K = 3$	0.984 (0.022)	0.741 (0.033)
Latent class, $K = 8$	0.973 (0.028)	0.761 (0.024)
Hierarchical Bayes	0.871 (0.021)	0.800 (0.020)

Notes: Means over 10 splits, standard errors in parentheses. Hit rate: share of respondents whose highest-predicted hidden bag is among their top-rated hidden bags.

5.2 Number of segments

The criteria disagree (Table 4, Figure 1). On raw ratings, LC partly separates respondents by their average rating, so I use centered ratings. With centered ratings BIC falls up to $K = 8$ and is almost flat from $K = 6$, although from $K = 7$ the smallest segment holds about 2% of respondents. Holdout prediction improves up to $K = 3$ and then stays flat, and eight segments still predict worse than HB. Only two segments are stable (ARI 0.84). LC and k-means partitions agree weakly (ARI 0.16-0.29), and k-means silhouette scores are about 0.2. Preferences in these data look closer to a continuous distribution than to a few groups.

Table 4. Choosing the number of segments

	Latent class fit				Holdout	Stability (ARI)	
K	BIC, raw	BIC	ICL	Smallest segment	RMSE	LC	k-means
1	7418	6652	6652	1.00	1.059 (0.032)	-	-
2	7186	6379	6399	0.48	0.993 (0.025)	0.84	0.80
3	6988	6335	6369	0.24	0.984 (0.022)	0.54	0.63
4	6921	6297	6337	0.18	0.978 (0.023)	0.43	0.50
5	6904	6265	6311	0.11	0.983 (0.026)	0.45	0.55
6	6900	6252	6294	0.11	0.983 (0.030)	-	-
7	6900	6252	6299	0.02	0.973 (0.025)	-	-
8	6899	6247	6290	0.02	0.973 (0.028)	-	-

Notes: BIC and ICL on centered ratings unless marked raw; $N = 122$. Holdout RMSE as in Table 3 (HB: 0.871). Stability: mean ARI between the full-sample partition and 50 bootstrap re-estimates.

The three-segment solution is still easy to read (Table 5): a price-sensitive segment that wants a large bag with a strap pad (24%), a segment that does not mind the colorful exterior and cares most about the laptop compartment (29%), and a less price-sensitive segment that strongly dislikes the colorful exterior (47%).

5.3 Revenue

In sample, every strategy with more than one bag beats the mass market (last column of Table 6). Out of sample this changes (Figure 2). Targeting LC segments lowers revenue by 1.5-3.9%, although with known preferences it would raise it by 1-7%. Targeting k-means segments gives about zero, and offering the segment bags to everyone gives at most 2%, which is within noise. The optimal line gains 2.6% with two bags and 7.2% with four, about half of its potential. The mass-market bag itself loses about 4% when chosen on half of the sample.

Targeting loses for two reasons. Each segment's bag is chosen from only 10-30 training respondents, and new respondents are often put into the wrong segment because the segments are unstable. The optimal line avoids both

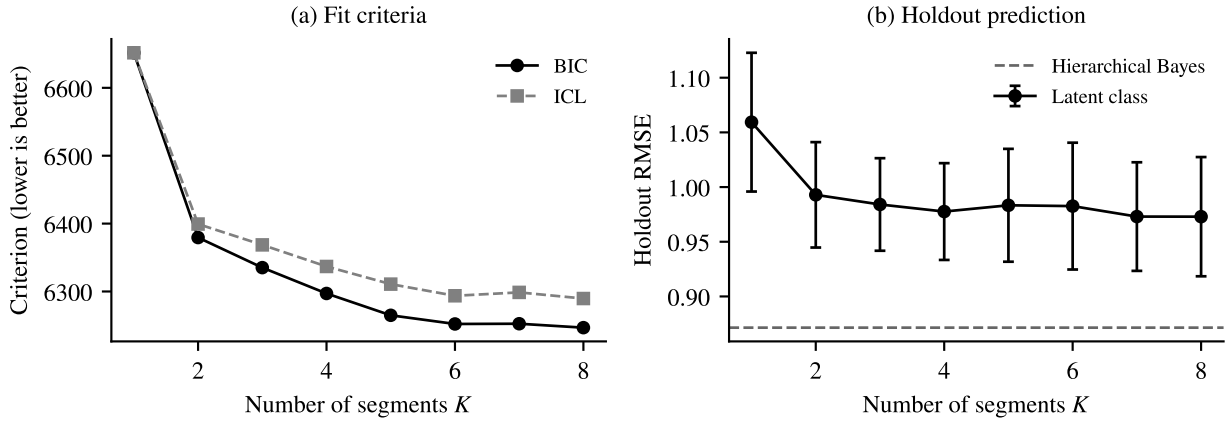


Figure 1. Fit criteria and holdout prediction by number of latent class segments. Error bars: 95% intervals across splits.

Table 5. Latent class segments, $K = 3$

	Segment 1	Segment 2	Segment 3
Price ($120 → $180)	−0.99	−0.74	−0.47
Exterior: reflective	−0.12	−0.37	−0.38
Exterior: colorful	−0.69	0.04	−1.94
Exterior: blue	−0.16	−0.29	−0.21
Size: large	0.77	−0.08	0.23
Strap pad	0.90	0.55	0.29
Water bottle pocket	0.68	0.65	0.20
Interior: file divider	0.37	0.56	0.33
Interior: padded laptop	0.61	0.89	0.46
Share of respondents	0.24	0.29	0.47

Notes: Partworths from LC on centered ratings (no intercept).

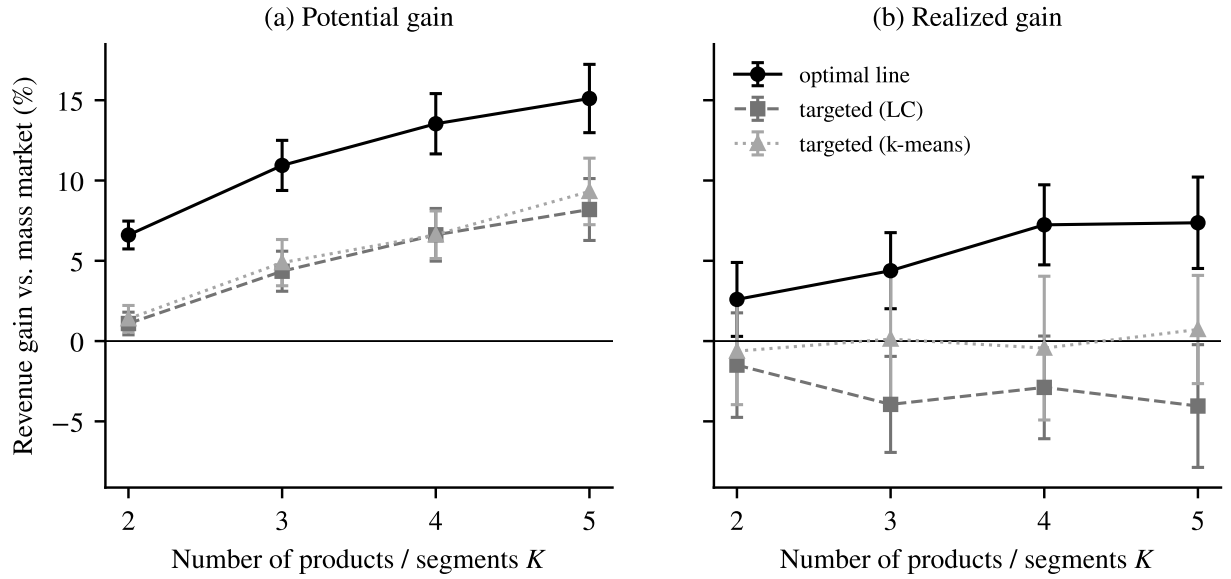


Figure 2. Out-of-sample revenue gain relative to one mass-market bag. Potential: bags chosen with the test respondents' own partworths. Realized: bags chosen on the other half. Error bars: 95% intervals across folds.

Table 6. Revenue gain relative to the mass market (%)

Strategy	K	Realized	Potential	In sample
Mass market ($ per consumer)		132.3	137.8	137.2
Optimal product line	2	2.6 (1.2)	6.6 (0.4)	4.8
	3	4.4 (1.2)	10.9 (0.8)	8.6
	4	7.2 (1.3)	13.5 (1.0)	10.6
Targeted segments, LC	2	−1.5 (1.7)	1.1 (0.4)	0.0
	3	−3.9 (1.5)	4.4 (0.6)	2.6
	4	−2.9 (1.6)	6.6 (0.8)	3.2
Targeted segments, *k*-means	2	−0.6 (1.7)	1.4 (0.4)	-
	3	0.1 (2.1)	4.9 (0.7)	-
	4	−0.4 (2.3)	6.6 (0.8)	-
Segment products, open, LC	2	0.5 (2.0)	−0.1 (0.9)	−2.9
	3	0.9 (1.9)	1.3 (1.4)	2.6
	4	1.5 (2.1)	1.9 (1.4)	1.7
Segment products, open, *k*-means	2	2.1 (1.7)	1.1 (0.6)	-
	3	−0.5 (2.1)	0.5 (1.4)	-
	4	2.2 (2.1)	4.0 (1.1)	-

Notes: Means over 20 folds of repeated 2-fold cross-validation, standard errors in parentheses. First row: revenue per consumer in dollars. In sample: bags chosen and evaluated on all 122 respondents.

problems: it is chosen on all training respondents, and consumers pick the bag they like.

6 Simulation

Design. Partworths are drawn as

$$\beta_i = \mu + s\, d_{k(i)} + \sqrt{w}\,\omega_i, \qquad \omega_i \sim N(0, \Sigma_W), \qquad \Sigma_W = \Sigma - \textstyle\sum_k \pi_k d_k d_k', \tag{3}$$

where μ and Σ are the HB estimates, d_k are the deviations of the three LC segments from the mean and π_k their shares. At the calibrated point segments explain 27% of the heterogeneity. Ratings are $z_j'\beta_i$ plus noise with the HB error variance, rounded to the 1-5 scale. I use five structures, (s, w): none $(0,0)$, continuous $(0,1)$, calibrated $(1,1)$, strong segments $(2,1)$ and distinct segments $(2, 0.25)$, and four sample sizes $n \in \{60, 120, 300, 1000\}$, with 20 markets per cell. The firm surveys n respondents, estimates HB and segments, and chooses three bags. Revenue is computed on 1,000 new consumers with their true partworths; they are assigned to segments from their ratings.

Results. Without heterogeneity no strategy gains, and targeting loses at most 0.3% (Figure 3, Table 7). With 60 respondents targeting does not pay in any structure. In a market like the real one it gains 0-1% at $n = 120$ and 4-5% at $n = 1000$, below the optimal line. With strong segments it gets close to its potential from $n = 300$. The optimal line gains in every market with heterogeneity, already at $n = 60$, and keeps 70-90% of its potential in the continuous, calibrated and strong-segment markets. The exception is distinct, tight segments: there the line keeps much less of its potential, and from $n = 300$ targeting does as well. HB assumes normal preferences and blurs tight segments, while LC fits this structure.

At the calibrated point with $n = 60$, close to the training half in Section 5, the simulation gives the same ordering as the data: the line gains and targeting does not. The line gain is larger in the simulation (9% vs. 4%), because the simulation evaluates on the true partworths of 1,000 consumers and assumes normal preferences.

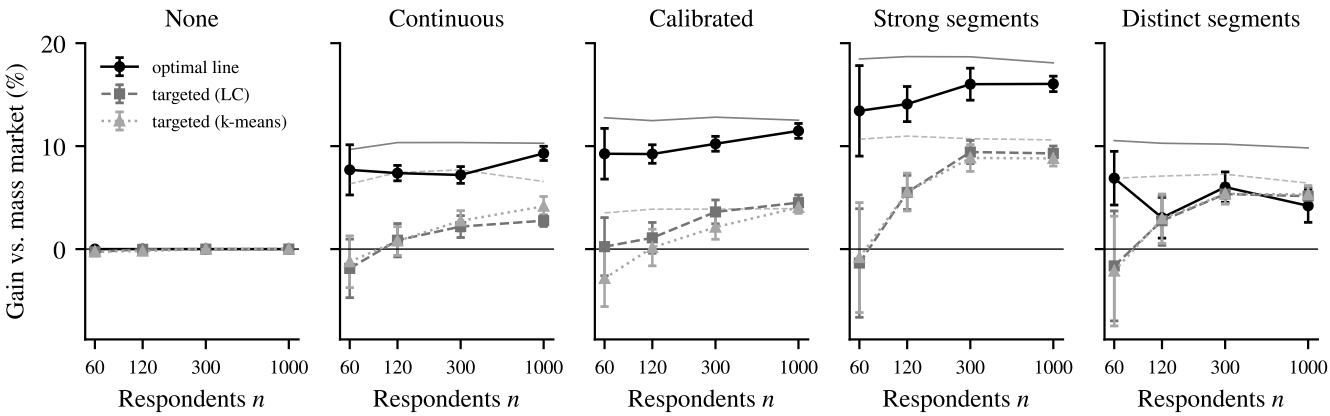


Figure 3. Realized gain of three bags relative to the mass market in simulated markets. Thin lines: potential gain of the line (solid) and of targeting (dashed). Error bars: 95% intervals over 20 markets.

Table 7. Simulation: revenue gain of three bags relative to the mass market (%)

		Optimal line		Targeted segments		
Structure	n	Realized	Potential	LC	k-means	Potential
None	60	0.0	0.0	−0.3	−0.2	0.0
	120	0.0	0.0	−0.0	−0.2	0.0
	300	0.0	0.0	−0.0	0.0	0.0
	1000	0.0	0.0	0.0	0.0	0.0
Continuous	60	7.7	9.7	−1.9	−1.2	6.3
	120	7.4	10.3	0.9	0.8	7.4
	300	7.2	10.3	2.2	2.7	7.7
	1000	9.3	10.3	2.8	4.2	6.6
Calibrated	60	9.3	12.8	0.2	−2.8	3.5
	120	9.2	12.5	1.1	0.2	3.9
	300	10.2	12.8	3.6	2.1	3.9
	1000	11.5	12.5	4.5	4.0	3.9
Strong segments	60	13.4	18.5	−1.4	−0.8	10.7
	120	14.1	18.7	5.5	5.5	11.0
	300	16.0	18.7	9.4	8.8	10.7
	1000	16.0	18.1	9.3	8.8	10.6
Distinct segments	60	6.9	10.5	−1.6	−2.1	6.9
	120	3.0	10.3	2.8	3.0	7.1
	300	6.0	10.2	5.3	5.4	7.3
	1000	4.2	9.8	5.1	5.3	6.4

Notes: Means over 20 markets per cell; the largest Monte Carlo standard error of a realized gain is 2.7 percentage points. Targeting potential uses k-means on true partworths and is a reference point, not an upper bound.

7 Can the firm tell whether to segment?

The firm does not know true preferences. I simulate 15 new markets per cell and compare rules that use only the firm's survey to choose between the mass market, three targeted LC segments and a three-bag line. The *BIC rule* targets segments if BIC prefers $K > 1$. The *in-sample rule* picks the strategy with the highest revenue on the survey. The *CV rule* picks the strategy with the highest revenue in 2-fold cross-validation within the survey. The *learned rule* is a ridge regression that predicts each strategy's gain from statistics the firm can compute (sample size, BIC gain, entropy, split-half stability of the segments, in-sample and cross-validated gains); it is trained on four structures and tested on the fifth. Regret is the revenue lost relative to the best of the three strategies.

Table 8. Regret of decision rules (% of the best strategy's revenue)

Rule	All markets		Mean regret by structure			
	Mean (SE)	Share $< 0.5\%$	Contin-uous	Cali-brated	Strong segments	Distinct segments
Always mass market	7.11 (0.29)	0.21	7.0	9.2	13.2	6.1
Always target segments	4.56 (0.25)	0.31	5.7	7.2	7.4	2.5
Target if BIC selects $K > 1$	4.54 (0.25)	0.32	5.7	7.2	7.4	2.5
Best in-sample revenue	0.66 (0.13)	0.87	0.6	0.8	0.0	1.8
Best cross-validated revenue	1.05 (0.15)	0.81	1.5	0.7	1.7	1.3
Learned rule (leave one structure out)	0.44 (0.09)	0.89	0.1	0.0	0.0	2.0
Always product line	0.43 (0.09)	0.90	0.1	0.0	0.0	2.0

Notes: 300 markets (5 structures $\times$ 4 sample sizes $\times$ 15). Without heterogeneity all rules have regret close to zero (not shown).

BIC prefers several segments in almost every market with heterogeneity, so the BIC rule nearly always targets and loses 4.5% of revenue (Table 8, Figure 4). The in-sample rule loses 0.7% and the CV rule 1.1%. The CV rule is worse at small n, where each half of the survey is too small, and the two are equal at $n = 1000$. Since the line is the best strategy in almost all markets, always offering a line loses only 0.4%. The learned rule does as well, but not better. The choice is hard only with distinct segments, where targeting is best in about half of the markets and the CV rule does best. Statistics that describe the segments predict the gain from targeting poorly (correlations of 0.2-0.4); split-half stability is the most useful of them.

On the real data BIC selects four segments, so the BIC rule recommends targeting, which loses about 4% out of sample. The other rules recommend the line.

8 Robustness

Demand rule. Table 9 repeats the comparison with a purchase threshold of 4 and with a smooth rule: the consumer picks the bag with the highest predicted rating and buys it with probability $\Phi((u-3)/\hat{\sigma})$, where $\hat{\sigma} = 0.67$ is the HB error standard deviation. I do not use a logit, because in a logit adding products raises the total purchase probability even when all consumers are identical. The main results hold. With threshold 4, open segment products do about as well as the line: when fewer people buy, having several bags matters more than which bags they are.

Replication. If segments describe stable types of consumers, a respondent should fall into the same segment when rating the same bags again. Table 10 shows that this happens much less often than bootstrap stability suggests: ARI is 0.38-0.49 for two segments and about 0.2 for three or four. The segment that strongly dislikes the colorful exterior appears in both measurements with almost the same partworth (-1.95 and -1.93); the other segment changes. The second measurement used the physical bags, so part of the difference may be a real change in evaluations. In either case, a firm that classifies a returning customer from new ratings would often put them in a different segment.

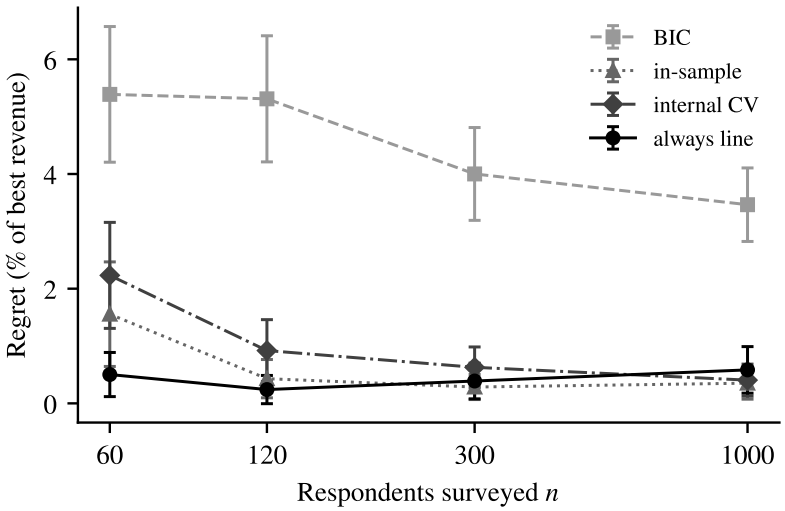


Figure 4. Regret of decision rules by sample size, averaged over structures. Error bars: 95% intervals.

Table 9. Revenue gain relative to the mass market under other demand rules (%)

		Threshold 3		Threshold 4		Smooth	
Strategy	K	Realized	Potential	Realized	Potential	Realized	Potential
Optimal product line	2	2.6	6.6	4.9	10.0	1.6	4.1
	3	4.4	10.9	7.8	16.5	3.6	6.8
	4	7.2	13.5	9.7	20.7	5.6	8.7
Targeted segments, LC	2	−1.5	1.1	2.3	2.6	−0.4	0.7
	3	−3.9	4.4	−2.5	8.1	−0.5	2.4
	4	−2.9	6.6	−2.0	12.2	−0.4	3.5
Targeted segments, k-means	2	−0.6	1.4	−1.9	2.2	−0.7	0.4
	3	0.1	4.9	−2.0	6.8	1.2	2.9
	4	−0.4	6.6	−2.1	11.9	1.5	4.3
Segment products, open, LC	2	0.5	−0.1	2.9	1.9	−1.3	−0.3
	3	0.9	1.3	6.8	8.5	−0.3	1.6
	4	1.5	1.9	9.4	9.9	−0.4	2.5

Notes: Means over 20 folds of repeated 2-fold cross-validation, as in Table 6.

Table 10. Agreement of segments between the two measurements (ARI)

K	LC, estimated separately	Offline ratings, online model	k-means on HB partworths	Bootstrap, online only
2	0.38	0.49	0.14	0.84
3	0.23	0.25	0.16	0.54
4	0.19	0.18	0.13	0.43

9 Conclusion

In these data, segmenting consumers and designing a bag for each segment does not increase revenue out of sample, even though it does in sample and even though BIC favours many segments. A product line chosen directly from individual preferences does increase revenue. The simulation shows when this changes: targeting needs well separated segments and a survey of several hundred people. For a firm, the practical rule is to decide on segmentation by revenue on held-out respondents, not by the fit of the segmentation model.

The study has clear limits. The data are stated ratings, purchase is defined by a threshold, and there are no costs or competitors, so the revenue numbers are illustrative. Out-of-sample revenue in the data is computed on estimated preferences. The simulation is calibrated to the same data and assumes normal heterogeneity within segments. The line is chosen on HB estimates, which work poorly when segments are tight; choosing it on mixture-based estimates is a natural next step.

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